Contextualizing Spotify's Global Hits

by Twisha Santra and Hayley Yoshikawa

Summary of Research Questions

1. **Which genres produce the most viral content? Does virality translate to sustained streaming growth or just short-term spikes?**

UK Garage, Synth-Pop, and Alternative Pop had the highest average viral scores among the genres in the dataset. As for genre's relationship with virality, there is no significant relationship between the two. However, there are differences in streaming volume and momentum across genres. So, genre is powerful in predicting a song's long-term streaming success. Virality is genre-blind and more short-term.

2. **For artists with multiple viral songs, is their music's viral score consistent across all viral tracks, or does it vary from track to track?**

Viral scores vary across artists, in a statistically significant fashion, but consistency differs by artist. The artists with the highest median score include Olivia Dean, Bad Bunny, and Zara Larsson, among others. Artists like Ed Sheeran, Rihanna, and Kendrick Lamar show relatively stable viral scores, whereas Taylor Swift or Olivia Dean show much more variability in their viral tracks. We concluded that virality is very much artist-dependent.

3. **Does the duration of a song's viral run depend on the artist's country of origin? For example, are British artists more likely to sustain their vitality than American ones?**

Country of origin is a statistically significant factor in both chart position and trending duration. Exogenous factors not included in the dataset that contribute to a song's virality, such as fanbase size or streaming infrastructure, vary by country. American artists consistently produce viral music, with songs appearing in all 3 longevity categories. Evergreen songs are concentrated across countries such as Canada, Great Britain, and Australia. By comparison, the shortest-duration new songs on the charts originate from countries such as South Korea and Norway.

Motivation

Music is an integral part of so many individuals' lives, including ours. Many people rely on music to exercise creative freedom, and some even make money from it. The music industry captures the experiences and perspectives of so many diverse communities. How we see music is, more so, something that can be used as a catalyst for connection. We were interested in questions about what drives a track's virality, ranging from individual artists to the countries of origin. Overall, we hope to gain insight into the factors that contribute to virality in the music industry. Applying those insights on a larger scale might help us approach connection with a more nuanced understanding of the origins of the music we consume.

Dataset

Spotify Global Trends & Popularity Analysis

This is a music streaming dataset from kaggle.com that explores the performance and metadata of songs on Spotify's global charts. This dataset explores the relationship among social virality, chart position, and streaming momentum across genres and artists' countries of origin. It is well-suited for looking into chart dynamics, viral momentum, genre behavior, and artist-origin patterns.

The dataset consists of 13 columns, including the track name, artist, streams, stream change, streams over 7 days, genre, artist's origin country, position (in ranking), days of trending, viral score, trend (rising or falling), popularity category, and longevity. In total, the dataset contains 7 string and 6 integer columns.

The exact collection period and whether the data reflects a single point in time or is aggregated across multiple dates are not specified by the dataset publisher, which is a limitation to keep in mind when interpreting trends over time.

Methodology

Research Question 1:

To answer the question, we will use the **genre**, **days**, **longevity**, and **viral_score** columns to determine whether there is a relationship between genre and virality strength. We plan to follow these steps:

1) Data Cleaning and Grouping: To predict with our models, it is important to identify sustained streaming growth from short-term spikes. To do so, we would need to explore the aforementioned columns and follow a typical cleaning process, dropping any NaN values and so on.
    a) Replace odd “country” values appropriately. For example, we replaced “Florida” with “US.”
    b) Remove rows with odd “genre” values.
2) Train ML Models: Using the processed data, we will build on the machine learning foundation developed in class to train an SVM model.
3) Evaluate Models: After training the model, we will compare its accuracy using the test set’s accuracy.
4) Prediction: Make predictions around a song’s long-term stream levels after its ongoing viral event based on genres, stream_change, and viral_score.
5) Descriptive Statistics and Calculations: We will calculate the mean, median, and standard deviation of viral scores across each genre.
6) Statistical Testing: Using a statistical significance test, we will determine whether viral scores differ significantly across genres.
7) Visualization: Segment by genre, to create box plots, swarm plots, etc.
8) Interpret Findings: Conclude whether virality depends on genre.

Research Question 2:

To answer the question, we will use the **artist** and **viral_score** columns, as their presence in the dataset indicates that the song in question was trending at some point. We plan to follow these steps:

1) Data Cleaning and Grouping: Explore the artist and viral score columns, addressing any NaN values and other anomalies. We also believe that it is important to only consider artists with multiple (2+) viral tracks so that we can apply descriptive statistics to a larger subset of data.
2) Descriptive Statistics and Calculations: We will calculate the mean, median, and standard deviation of viral scores across each artist’s tracks.
3) Statistical Testing: Using a statistical significance test, such as ANOVA, we will determine whether differences in viral scores across a particular artist’s tracks are significant.
4) Visualization: Segment by artist, to create box plots, scatter plots, etc.

5) Interpret Findings: Conclude if virality is a consistent trait of an artist or if it varies track to track.

Research Question 3:

To answer the question, we will use the **days** column as our measure of viral run duration and the country column as our measure of artist origin. We also plan to follow these steps:

1) Data cleaning and Grouping: Examining the country column to identify unique origin countries presented in the dataset. We will likely set a minimum threshold for observations per country and exclude countries with fewer than that number of tracks from the analysis.
    a) Replace inconsistent "country" values appropriately. For example, change the "Florida" entry to "US" to match other values.
2) Descriptive Statistics and Calculations: We will calculate the mean, median, and standard deviation of the days column. Because we anticipate right-skewed distributions, the median will be the primary measure.
3) Statistical Testing: Using a statistical significance test, like ANOVA, we will determine whether or not there is a significant relationship between the artist's country of origin and the duration of a song's viral run.
4) Visualization: Segment by country of origin to create box plots, scatter plots, etc.
5) Interpret Findings: Conclude if the duration of a song's viral run depends on the artist's country of origin.

EDA Methodology

1) Visually and computationally inspected the Spotify Global Trends dataset to evaluate if there were missing values and the total number of rows and columns.
2) Identify variables of interest and summarize them. For quantitative variables, conduct a seven-number (mean, standard deviation, minimum, first quartile, median, third quartile, maximum). For categorical variables, generate a list of all unique values that a specific variable can take and count them.
3) Generate 2 exploratory visualizations relating to our research questions and analyze them.
4) Determine the appropriate statistical test for each research question and verify that our variables of interest meet each test's requirements.
5) EDA Testing. See rationale in "Testing" section below.

EDA Summary

Dataset Basics: Size & Missing Values

Our Spotify Global Trends dataset has **178 rows** and **13 columns**. The rows of this dataset represent different trending tracks on Spotify. Furthermore, the columns of this dataset represent a variety of factors about these trending tracks. Some of these factors include the total number of streams, streams over 7 days, chart position, and so on. Using Pythonic functions .isna() and .sum(), we confirmed that there are no NaN values in any column, ensuring our dataset is completely clean and does not require imputation or row removal.

Research Questions & Variables of Interest

Research Question 1 (Genre & Virality):
This question examines whether particular genres are prone to sustaining long-term streaming or to producing only short-term spikes. Key variables are **genre, viral_score, streams, longevity, and stream_change.** Genres serve as the comparison category, while viral score measures overall success. Stream change helps identify temporary spikes, whereas the combination of streams and longevity provides insight into sustained popularity beyond an artist's existing fanbase.

Research Question 2 (Artist Consistency):
This question examines whether artists with multiple viral tracks maintain consistent virality scores. The artist is what we filter the dataset on, as we automatically filter to artists who appear more than once. Key variables are **artist_name, viral_score, popularity_category, and track_name** (serve mainly as identifiers).

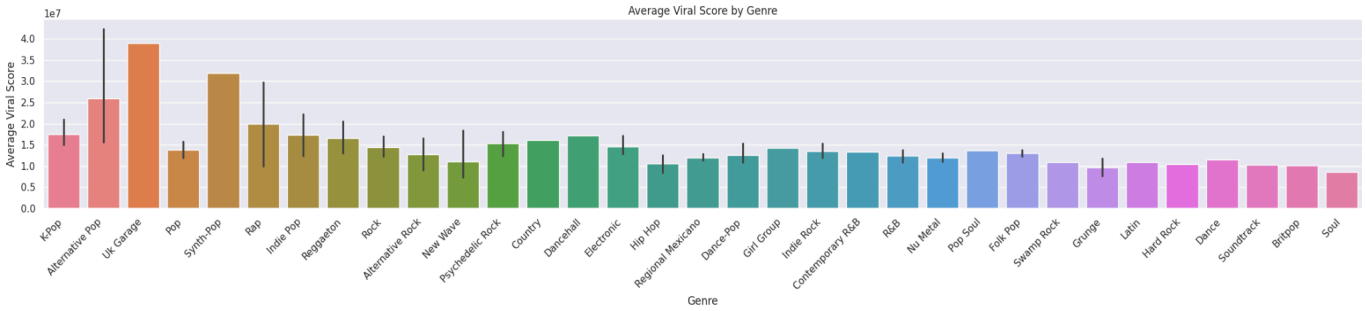
Research Question 3 (Country of Origin & Viral Duration):
This question tests whether there is a relationship between how long a song remains viral and the country of origin. Key variables are **country, days, longevity, and pos**. Countries serve as the grouping variable, while days and longevity measure the persistence of popularity. Chart position helps indicate the magnitude of a song's success.

Notable Observations:

- Two variables (trend and 7-day streams) are acknowledged but intentionally excluded from the three research questions, as they are less directly related to our core analysis.
- The popularity_category binary split (trending vs average) will be a useful lens for RQ2 but may limit nuance, and is worth keeping in mind during analysis.
- Variable summaries show that streams are right-skewed as they are driven by a small number of high-performing tracks, which reinforces why longevity is needed as a complement in RQ1
- Genre and Country are two primary categorical grouping variables, and their distributions in the dataset will affect how many tracks fall into each category comparison group.

Results

Visualization 1:



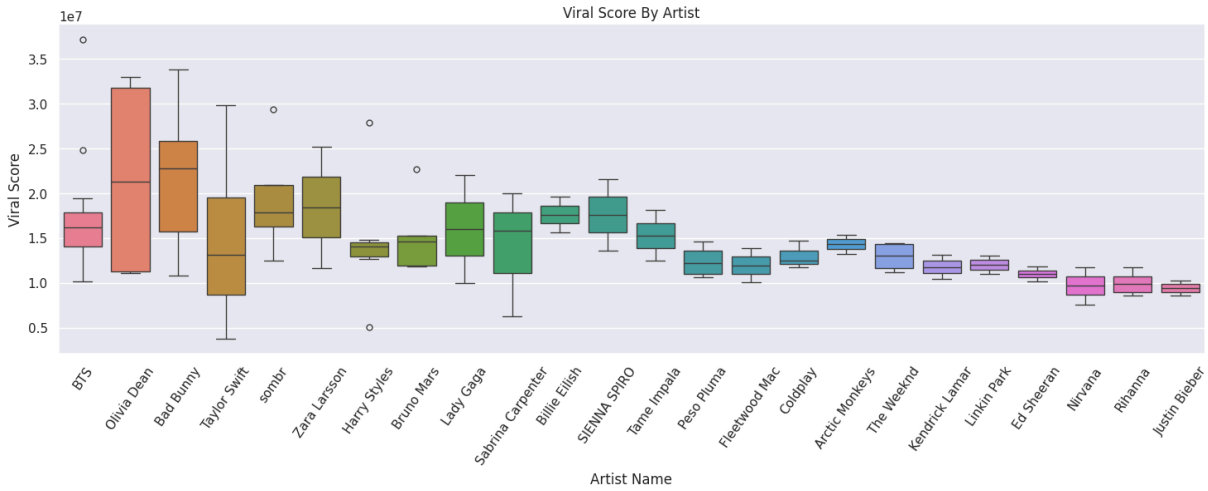
The plot shows the average viral score across Spotify's music genres, highlighting differences in how strongly virality varies by genre.

This plot shows the relationship between music genres and viral scores, which are quantitative measures of a song's short-term popularity and spread. We chose a bar plot because it provides a clear comparison of average viral performance across multiple genres, revealing which genres tend to produce more viral content overall. The mean is used as a summary statistic, and the error bars show variability and statistical uncertainty between genres.

From the plot, the genres UK Garage, Synth-Pop, and Alternative Pop appear to be the highest-performing, with average viral scores approaching 4.0×10^7. This is somewhat surprising, as UK Garage and Synth-Pop have historically been more niche genres. Yet, they outperform more mainstream categories like Pop, Hip-Hop, and R&B, which appear to fall in the middle of the distribution. One reason for this could be that a single hit, breakout song disproportionately inflates the genre's average, rather than reflecting consistent genre performance.

The error bars in this visualization support this interpretation, as genres like Rap and Indie Pop display wide confidence intervals, indicating high variability and suggesting that viral scores are not consistent, likely because they are influenced by a few highly successful songs rather than by the performance of all tracks in that genre. An important implication of these findings is that genre alone is an incomplete predictor of virality. To fully answer this question, it is also necessary to examine longevity and stream change.

Visualization 2:



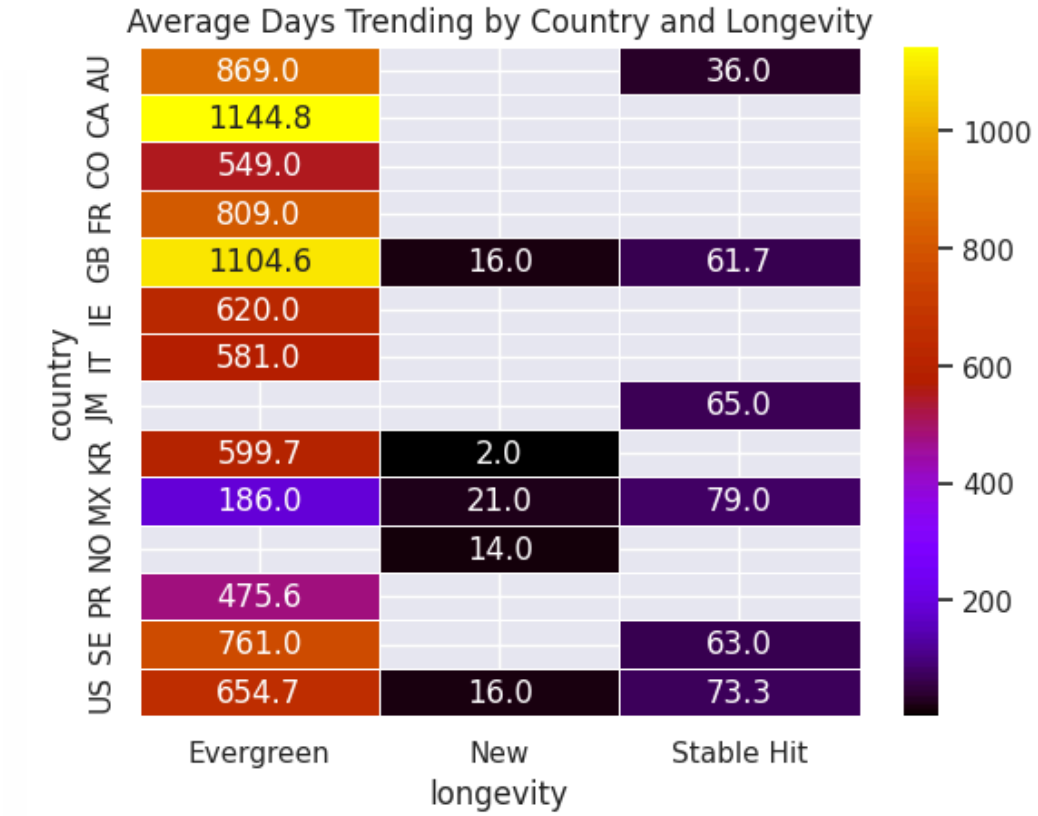
Median viral scores for artists with 2+ viral songs range from 1.5e7 to 2e7, which is higher than the dataset median of ~1.26e7.

This plot displays the distribution of viral scores among artists with 2 or more viral songs (i.e., those who appear more than once in the dataset). Relevant to research question 2, the above plot helps us determine if virality is consistent for certain artists or largely unpredictable. We chose a boxplot for this question because it portrays the overall distribution of data (median, spread, and outliers) for each artist's viral scores across their several viral tracks. As such, comparing artists represented in the dataset is very intuitive. Artists such as Ed Sheeran, Justin Bieber, Rihanna, Kendrick Lamar, and Arctic Monkeys have smaller interquartile ranges and overall ranges, indicating their viral scores are relatively stable across tracks. Alternatively, Taylor Swift, Dominic Fike, Olivia Dean, and Bad Bunny have much wider spreads. As a result, there is greater variability in how their recent popular songs have performed. The artists that

have the highest median viral score include Olivia Dean, Bad Bunny, Zara Larsson, sombr, and Billie Eilish.

Some of the most interesting attributes of this visualization lie in the extreme virality of some artists. Some immediate standouts include Olivia Dean and Bad Bunny, the former being an up-and-coming artist. Both artists' stories are fairly unique. Bad Bunny has a track record for success and has established his footing in the music industry for 10 years now. Standout events such as winning the Grammy for Album of the Year for DeBÍ TiRAR MáS FOToS and performing at the 2026 Super Bowl Halftime show only further propel his virality. Olivia Dean's fame is rooted in social media and the traction she gains from influencers using her music in their posts. Her music is also generally cross-cutting across generations. Many critics have regarded her discography as timeless. However, because she is a fairly new artist, she lacks the dedicated fans that apply upward pressure on her streams. This means that, beyond songs trending on social media, Olivia Dean will struggle to earn streams and remain on the charts, perhaps explaining the large variability in her viral score across charting songs. The broader implication of this plot is that sustained virality stems from different sources depending on the artist. For established artists, virality is reinforced by institutional recognition and cultural events. For emerging artists, virality is often socially driven, making it less stable. If we wanted to classify an arbitrary song as likely to go viral (or not), it is important to consider the larger ecosystem surrounding the artist.

Visualization 3:



Average days trending for Evergreen tracks range from 186.0 to 1144.8 days across countries, while Stable Hit tracks range from 36.0 to 79.0 days, and New tracks range from 2.0 to 21.0 days. This highlights the stark difference in chart persistence within and across longevity categories.

The heatmap of average days trending by Country and Longevity category shows each country's average duration of songs' viral runs. Longevity is grouped into three categories: Evergreen (timeless tracks that remain relevant over long periods), New (recently added, fresh content that is trending), and Stable Hit (established, consistent hits that are often long-running tracks). Countries will also differ considerably in which categories their songs fall into and how long these songs trend within each category.

The heatmap reveals some interesting differences between countries. CA (Canada) and GB (Great Britain) lead among the Evergreen tracks, with average trending of 1144.8 and 1104.6 days, respectively. Both of which have substantially more days trending than countries like MX (Mexico), at 186.0 days, and PR (Puerto Rico), at 475.6 days. CA and GB's dominance in the Evergreen category suggests that these countries have a high level of cultural longevity and maintain relevance in the charts for years rather than months. This could potentially reflect the global reach of English-language music, as both countries have mainly English-speaking artists.

For the Stable Hit category, MX (Mexico) leads with an average of 79.0 days, followed by the US (United States) with an average of 73.3 days. It is interesting to note that these countries outperform GB at 61.7 days despite GB's dominance in the Evergreen category. This suggests that GB produces more songs with long-term staying power, but countries like MX and JM are more consistent at producing tracks that sustain chart positions. Overall, these results support the idea that country of origin relates to viral duration through factors such as language, genre, and platform performance promotion.

Challenge Goals:

Result Validity

For all three research questions, we examined the relationship between a categorical independent variable and a continuous dependent variable across multiple groups. Under these conditions, the **One-Way ANOVA Test** is the most appropriate statistical test. However, ANOVA assumes normality, and the dependent variable must be normally distributed within groups. To assess these assumptions, we checked histograms of the dependent variables (viral_score and days). Both variables exhibited right-skewed distributions, violating the normality assumption. Because these distributions did not satisfy the normality assumption for ANOVA, we selected the **Kruskal-Wallis Test** (a nonparametric equivalent of the one-way ANOVA), which does not assume normality.

We also wanted to examine the relationship between two categorical independent variables. As such, we used the **Chi-Squared Independence Test**. For this particular test, the only assumption that needs to be fulfilled is that there are sufficiently large sample sizes, which means the expected frequency in every cell of a constructed contingency table is 5. We accounted for this in our implementation of the statistical tests.

Research Question 1:

Null Hypothesis 1 ($H0_1$): *Viral scores, streams, and stream changes do not differ across genres. In other words, there is no significant difference in viral scores across music genres.*

- Independent Variable: genre
- Dependent Variable: viral_score, streams, stream_change

- Statistical Test: **Kruskal-Wallis Test**

```
~~~ Statistical Test for Research Question 1 ~~~

 -- Kruskal-Wallis Test for viral_score by Genre --

The Kruskal-Wallis H statistic is 43.23681589081275
The p-value is 0.08872892747421438
We fail to reject the null hypothesis.
------------------------------------------------
 -- Kruskal-Wallis Test for streams by Genre --

The Kruskal-Wallis H statistic is 56.57890136324795
The p-value is 0.004695896057369627
We can reject the null hypothesis!
------------------------------------------------
 -- Kruskal-Wallis Test for stream_change by Genre --

The Kruskal-Wallis H statistic is 76.01339918480039
The p-value is 1.9166865533595858e-05
We can reject the null hypothesis!
------------------------------------------------
```

Findings

We were able to reject the null hypothesis for streams versus genre and stream_change versus genre. However, we were unable to reject the null hypothesis for the viral_score versus genre comparison. The genre of a song affects streaming volume and momentum, but not directly virality. Genre plays a larger role in shaping long-term streaming behavior, whereas virality exists independently of genre and on a shorter timeline.

Null Hypothesis ($H0_2$): *Longevity is independent of genres.*

- Independent Variable: genre
- Dependent Variable: longevity
- Statistical Test: **Chi-Squared Test of Independence**

```
 -- Chi-Square Test of Independence for longevity by Genre

The Chi-Squared statistic is 145.9409588192052
The p-value is 2.426794645440987e-08
We can reject the null hypothesis!
Warning: some expected frequencies are below 5.
------------------------------------------------
```

Findings

We were able to reject the null hypothesis for longevity versus genre, which suggests that the longevity classification of a song does depend on its genre. However, the expected frequency assumption of the Chi-Squared test is violated, which calls into question the reliability of our conclusion. Positively, the chi-squared statistic is large, and the p-value is extremely small, so our findings seem robust. The assumption violation, though, still warrants caution.

Research Question 2:

Null Hypothesis (H0): *Viral scores do not differ across an artist's viral tracks. In other words, there is no significant difference in viral scores across tracks for artists with multiple viral songs*

- Independent Variable: artist_name
- Dependent Variable: viral_score
- Statistical Test: **Kruskal-Wallis Test**

```
------------------------------------------------
 ~~~ Statistical Testing for Research Question 2 ~~~

 -- Kruskal-Wallis Test for viral_score by Artist Name --

The Kruskal-Wallis H statistic is 195.64818581831346
The p-value is 2.445797136323088e-29
We can reject the null hypothesis!
```

Findings

We were able to reject the null hypothesis for viral_score versus artist, indicating that viral scores differ statistically across artists. Whether an artist produces consistently viral content or experiences isolated spikes varies from artist to artist. As such, our second visualization is a better fit for this particular aspect of the question.

Null Hypothesis ($H0_2$): *Popularity category is independent of artist name.*

- Independent Variable: artist
- Dependent Variable: popularity_category
- Statistical Test: **Chi-Squared Test of Independence**

```
 -- Chi-Square Test of Independence for popularity_category by Artist Name--

The Chi-Squared statistic is 93.10877192982457
The p-value is 2.125587801464139e-10
We can reject the null hypothesis!
Warning: some expected frequencies are below 5.
-----------------------------------------------
```

Findings

We were able to reject the null hypothesis for popularity_category versus artists, suggesting that a song's popularity classification depends on its genre. This means that whether an artist is considered trending or average depends on the artist. However, the expected frequency assumption of the Chi-Squared test in this scenario is also violated, warranting caution in our conclusion.

Research Question 3:

Null Hypothesis (H0): *There is no significant difference in the duration of a song's viral run based on the artist's country of origin.*

- Independent Variable: country
- Dependent Variable: days (on viral charts), pos (on viral charts)
- Statistical Test: **Kruskal-Wallis Test**

```
-----------------------------------------------
 ~~~ Statistical Testing for Research Question 3 ~~~

 -- Kruskal-Wallis Test for days by Country --

The Kruskal-Wallis H statistic is 44.40470924317085
The p-value is 2.629199127986356e-05
We can reject the null hypothesis!
-----------------------------------------------
 -- Kruskal-Wallis Test for pos by Country --

The Kruskal-Wallis H statistic is 38.81223562814853
The p-value is 0.00021421674319809018
We can reject the null hypothesis!
-----------------------------------------------
```

Findings

We were able to reject the null hypothesis for both days versus country and pos versus country. This suggests that there is a statistically significant difference between a song's days of trending and position on the charts as compared to its country of origin. Some countries consistently produce tracks that chart higher and remain there longer. As such, a song's country of origin is a factor in global streaming performance.

Null Hypothesis ($H0_2$): *Longevity category is independent of a song's country of origin.*

- Independent Variable: country
- Dependent Variable: longevity
- Statistical Test: **Chi-Squared Test of Independence**

```
 -- Chi-Square Test of Independence for longevity by Country

The Chi-Squared statistic is 103.56185949080941
The p-value is 3.2578884810473207e-11
We can reject the null hypothesis!
Warning: some expected frequencies are below 5.
-----------------------------------------------
```

Findings

We were able to reject the null hypothesis for the country-by-longevity interaction, suggesting that a song's longevity classification depends on its country of origin. This suggests that songs from certain countries are more likely to achieve long-term chart presence than those from others. However, the expected frequency assumption of the Chi-Squared test in this scenario is also violated, warranting caution in our conclusion.

Machine Learning (RQ1)

When we chose the research questions and the data set for this project, we wanted it to address aspects of machine learning, which is best applied to complex situations. Music virality is a highly nuanced outcome influenced by many measurable factors (such as genres, streams, and engagement patterns), some of which are almost inevitably outside the dataset we chose. Due to this complexity, we believe more sophisticated predictive modeling could reveal patterns that simpler data operations might miss. To that end, we built on the materials we learned in class to train a **Support Vector Machine (SVM)** model to predict a song's longevity classification (Evergreen, New, or Stable Hit) based on genre, viral_score, and stream_change. SVM was chosen because it performs well on small datasets and is well-suited to multi-class classification, which is exactly what we need for predicting longevity categories. The variables were chosen because our earlier statistical analysis identified a meaningful relationship between genre and stream_change. Viral score was included as a feature, despite showing no statistically significant relationship with genre, because it likely still carries a predictive signal to some extent. Since we are working to predict a song's long-term virality, we excluded the streams variable and focused on stream_change to better capture a song's momentum (whether it is gaining or losing traction over time). We thought this would be a much stronger indicator or sustained chart presence than raw streams alone.

Model Evaluation:

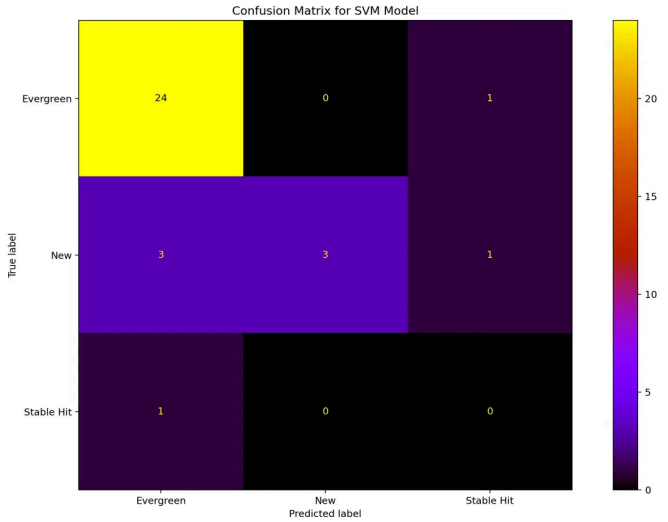
```
Accuracy: 0.8181818181818182
              precision    recall  f1-score   support

   Evergreen       0.86      0.96      0.91        25
         New       1.00      0.43      0.60         7
  Stable Hit       0.00      0.00      0.00         1

    accuracy                           0.82        33
   macro avg       0.62      0.46      0.50        33
weighted avg       0.86      0.82      0.81        33
```

The model achieved an overall accuracy of 82%, performing the strongest on Evergreen tracks. The F1-score for this category was 0.91. On the other hand, this model struggled with underrepresented classes. Stable Hit, for example, had only one instance in the test set and also had the lowest F1-score. SVM models are known to perform poorly on imbalanced data sets, so to cover our bases, training multiple models with different hyperparameters will help capture trends in our data. This would be a meaningful direction for future engagement we might pursue for this particular line of questioning.

Model Visualization:



Confusion matrix for the SVM model that compares the predicted label to the true label. This model performs the best on Evergreen tracks. For the other two categories, SVM performs rather poorly but better in accurately classifying New tracks. Every track in the Stable Hit category was misclassified.

Impact and Limitations

The findings from the analysis we have conducted for this project have direct applicability to the music industry and to music fans more broadly. For example, executives at record labels might use our conclusions to identify new artists to sign. If record labels are looking for long-term stability, they might choose to sign Canadian artists. If they are looking for a genre-based approach, they might choose to take on a UK Garage musician. From a musician's standpoint, our findings around what genres and viral artists might be very insightful. A new artist trying to find their own sound can draw inspiration from specific genres or artists based on their viral preferences. For example, if an individual prioritizes more stable virality, they might refer to Justin Bieber's or Kendrick Lamar’s music.

However, a significant limitation within our analysis is that it inadvertently excludes and misrepresents certain groups. The dataset is heavily skewed toward artists from English-speaking countries. As a result, insights related to a song’s country of origin are not generalizable to the global music landscape. This dataset is not representative of the rich music scenes in underrepresented countries because, most often, they are cultivated by streaming platforms other than Spotify. As a result, artists from underrepresented countries appear far less frequently in the dataset, in turn, limiting our ability to draw meaningful conclusions about their performance. The geographic bias clearly present in Spotify could reinforce existing blind spots, causing actors to overlook underrepresented markets that are not actually underperforming.

Within our data, there are also limitations, including its small size (only 178 rows), which contribute to imbalances in the SVM model we worked with and to expected-frequency violations in our chi-square independence tests. Further, the dataset’s period is unspecified and static. The data reflects a single, aggregated period. The music industry is very dynamic. For instance, the release of Olivia Rodrigo’s new

singles has most definitely shifted the charts between when this data was created and now. This limits the trends we can interpret with this data to a shorter horizon. Our analysis is more exploratory than wholly predictive. The patterns we identified are supported for this particular snapshot of Spotify data, but Spotify alone is not representative of the global music industry. Our conclusions are meant to be informative and should not be used to support large investment decisions in a music artist or other similar pursuits.

Work Plan Evaluation

The work plan we proposed previously was partially accurate in some areas, but we definitely underestimated how long some tasks would take. For task 1 (data cleaning and analysis), we were roughly on track, with about 4 hours of work time. However, we identified inconsistencies in the country and genre columns of our data that required further cleaning, including replacing irregular values to ensure consistency across the dataset. This added some unexpected time, but all issues have been resolved. For task 2 (tackling our research questions), we originally estimated 12 hours, which was a bit of an overestimation, as it took about 8 hours to complete the visualizations and analyze the three research questions. For task 3, we originally estimated 6 hours to finish the challenge goals, and we spent more time on this task than expected. Roughly 2 hours were spent determining the appropriate statistical tests, and the actual implementation of both challenge goals (ML modeling and result validity testing) took about 5-6 hours. Task 4 has taken by far the most time due to the level of written justification required for analytical decisions made throughout the project. Code cleanup and refactoring from a Jupyter notebook into Python modules added additional time, but has since been completed. Testing was conducted across our EDA process, and everything has been verified and finalized. With all tasks now complete, our project is ready for submission.

Testing

We decided to test with a smaller dataset (test_data), by creating a tiny dataframe where we know the answer and using assert statements to check the expected count. We wanted to confirm that our function (unique_values) could handle duplicates correctly, count them accurately, and print results without errors, regardless of the size of our larger dataset. We structured the test in three parts: first, checking that the function runs without errors, then verifying that duplicates are collapsed and counts match our known answers, and finally connecting back to spot_data. Because we manually defined the test data and their expected outputs, we can trust these results. If the function returns anything other than what we know is true, the assert statement should catch it and tell us what failed.

We also decided to test whether our function virality_score_by_artist() properly filters for artists with 2+ viral songs and accurately plots their viral scores. In our smaller dataset, only the artist Sombr appeared twice, so we expect his name to be the only value in the filtered artist_name column. We used assert statements to ensure Sombr was included in the filtered dataset, while the other artists were *excluded*. We also expect only one box plot (for sombr) in the visualization. This is exactly what our function produced.

Additional Sources

- Seaborn Heatmap
- Packing and Unpacking Arguments in Python
- Scipy kruskal (Kruskal-Wallis Test)
- Scipy chi_contingency (Chi-Squared Independence test)
- Pandas crosstab
- Chi-Squared Independence Test Python Explained
- Scikit-Learn StandardScaler
- Support Vector Machines (SVMs) Python Explained
- Machine Learning Confusion Matrix Visualization
- SVM Failures